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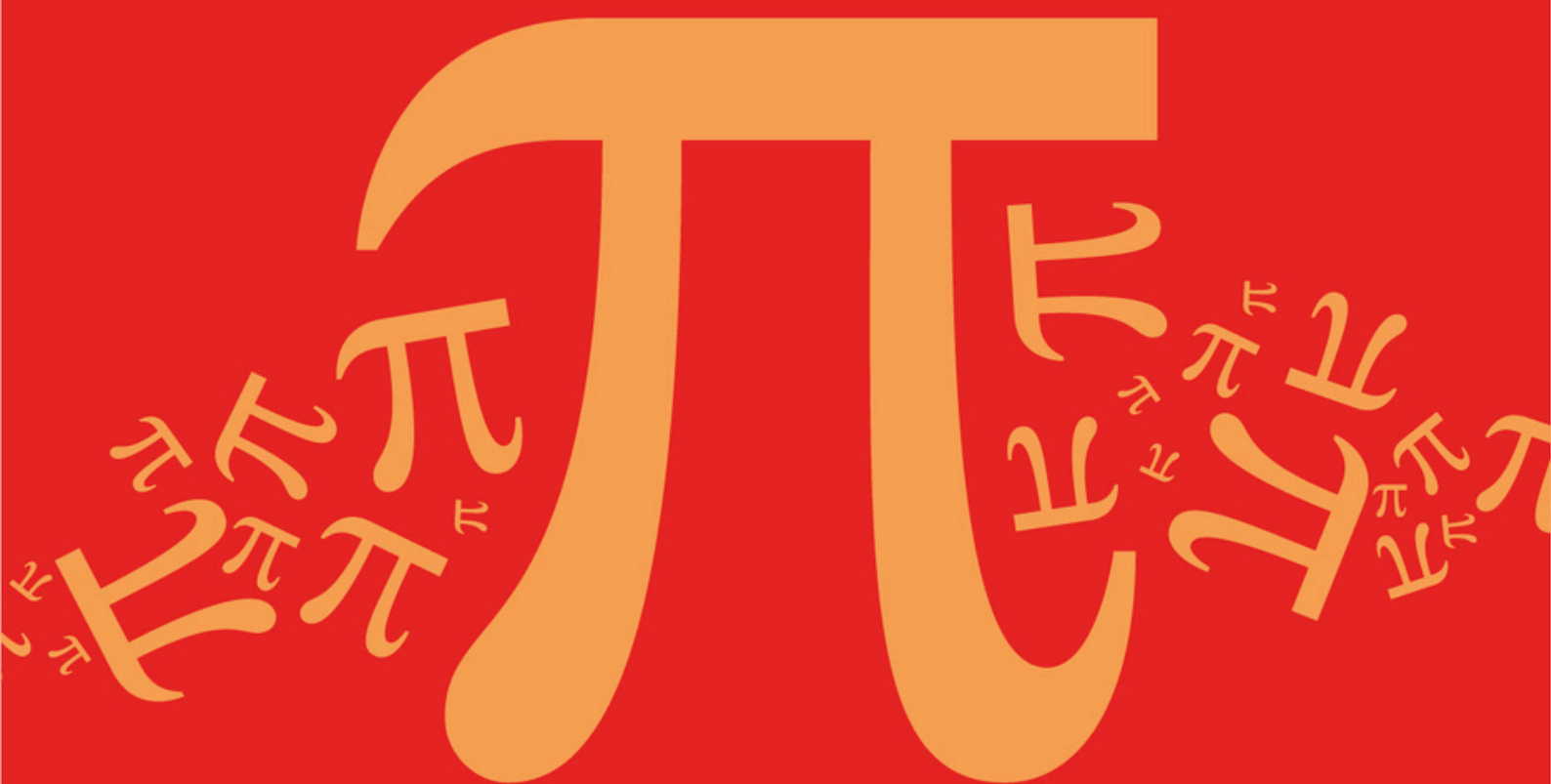
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Detailed mark scheme

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AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B3: Roots Occur in Conjugate Pairs

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

B: Complex Numbers

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B3: Roots Occur in Conjugate Pairs

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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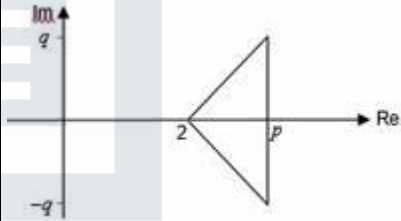
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Correctly identifies the complex conjugate as a root.	AO1.1b	B1	2 nd complex root is $2 + 3i$
	Forms one of the following equations (or their equivalents) $\alpha + \beta + 2 - 3i = 0$ or $\alpha\beta(2 - 3i) = \pm 52$ or $\alpha\beta + \beta(2 - 3i) + (2 - 3i)\beta = \pm m$ with an equation to find m . Or forms an equation to find m and then solves the cubic for their value of m .	AO1.1a	M1	sum of roots $= -\frac{b}{a}$ $\therefore \alpha + 2 + 3i + 2 - 3i = 0$
	Finds correct third root	AO1.1b	A1	$\alpha + 4 = 0$ $\alpha = -4$
				2 nd complex root is $2 + 3i$ product of roots $= -\frac{d}{a}$ $\therefore \alpha(2 + 3i)(2 - 3i) = -52$ $\alpha(4 - 9i^2) = -52$ $\alpha = \frac{-52}{13}$ $\alpha = -4$
(b)	Forms a correct equation to find m . May be seen in part (a).	AO1.1a	M1	$\therefore (-4)^3 + m(-4) + 52 = 0$
	Finds correct value of m .	AO1.1b	A1	$-64 - 4m + 52 = 0$ $-12 = 4m$ $m = -3$
				$\sum \alpha\beta = \frac{c}{a}$ $\therefore (2 - 3i)(2 + 3i) + (-4)(2 - 3i) + (-4)(2 + 3i) = m$ $4 + 9 - 8 + 12i - 8 - 12i = m$ $m = -3$
				Total 5 marks

Q2.

Marking	AO	Marks	Typical Solution
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Instructions			
Writes β and γ in the form $p \pm qi$ (seen anywhere in the solution)	AO2.5	B1	Real coefficients $\Rightarrow \beta = p + qi$ and $\gamma = p - qi$
Uses “sum of the roots = $-b/a$ ” together with a conjugate pair to determine the real part (p) of β and γ	AO3.1a	M1	$\alpha + \beta + \gamma = 8$ $\Rightarrow 2 + p + qi + p - qi = 8$ $\Rightarrow 2 + 2p = 8$ $\Rightarrow p = 3$
Uses ‘(their p)’ – 2 and the area of the triangle on an Argand diagram to determine the imaginary parts of β and γ	AO3.1a	M1	$(p - 2)q = 8$ $\Rightarrow q = 8$ 
Uses a correct method to find the value of c or d using ‘their’ values of $p \pm qi$	AO1.1a	M1	$\beta = 3 + 8i$ and $\gamma = 3 - 8i$
Obtains correct values for c and d . CAO	AO1.1b	A1	$d = -\alpha\beta\gamma = -146$ $c = \sum \alpha\beta = 85$
Total 5 marks			

Q3.

(a) (i) $(z - 2i)^* = (x + yi - 2i)^* = x + (2 - y)i$
 $x + 2i - yi$ OE rearrangement

B1

1

(ii) $(z - 2i)^* = 4iz + 3 = 4ix + 4i^2y + 3 = 4ix - 4y + 3$

$x + (2 - y)i = 4ix - 4y + 3$ (#)
 $i^2 = -1$ used

B1

Real parts: $x = -4y + 3$

Imaginary parts: $2 - y = 4x$

Attempting to equate, without mixing real and imaginary terms, **both** the real parts and the imag. parts for the c's eqn (#).

M1

If not corrected, ft on $[c's(a)(i)] = 4ix - 4y + 3$ provided both the resulting linear equations have non zero x, y and const terms

A1F

$$y = \frac{2}{3}, \quad x = \frac{1}{3}$$

Solving correct equations, to obtain either $x = \frac{1}{3}$ OE or $y = \frac{2}{3}$ OE

A1

$$(z =) \frac{1}{3} + \frac{2}{3}i$$

A1

5

- (b) (One of the) coefficients (of the quadratic equation is) not real.
 OE eg Sum of roots is $-10i$ so p cannot be real if roots are $p \pm qi$

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E1

1

[7]

Q4.

(a) (i) $\alpha + \beta + \gamma = 0$

B1

1

(ii) $\alpha\beta\gamma = -q$

B1

1

(b) $\alpha^3 + p\alpha + q = 0$

M1

$$\sum \alpha^3 + p \sum \alpha + 3q = 0$$

m1

$$\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

AG

A1

Alternative:

Use of $(\sum \alpha)^3 = (\sum \alpha^3) + 6\alpha\beta\gamma + 3(\sum \alpha \sum \alpha\beta - 3\alpha\beta\gamma)$

(M1)

Substitution of $\sum \alpha = 0$

(m1)

Result

(A1)

3

(c) (i) $\beta = 4 - 7i, \gamma = -8$

B1,B1

2

(ii) Attempt at either p or q

M1

$p = 1$

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A1F

$q = 520$

ft incorrect roots provided p and q are real

A1F

3

(d) Replace z by $\frac{1}{z}$ in cubic equation

or $\sum \frac{1}{\alpha} = -\frac{p}{q}, \sum \frac{1}{\alpha\beta} = 0, \frac{1}{\alpha\beta\gamma} = -\frac{1}{q}$

ft on incorrect p and / or q

M1

A1F

$520z^3 + z^2 + 1 = 0$ coefficients must be integers

CAO

A1

3

Q5.

(a) (i) $\beta = 2 - 2\sqrt{3}i$

B1

1

(ii) $\alpha\beta\gamma = -8$
Allow for +8 but not ± 16

M1

$\alpha\beta = 16$

B1

$\gamma = -\frac{1}{2}$

A1

3

(iii) Either $\frac{-p}{2} = \alpha + \beta + \gamma$
 or $\frac{q}{2} = \alpha\beta + \beta\gamma + \gamma\alpha$

SC if failure to divide by 2 throughout,
 allow M1A1 for either p or q correct ft

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$p = -7, q = 28$
ft incorrect γ

A1F, A1F

3

Alternative to (a)(ii) and (a)(iii)
 $(z^2 - 4z + 16)(az + b)$

(M1)

$\alpha\beta = 16$

(B1)

$a = 2, b = +1, \gamma = -\frac{1}{2}$

(A1)

Equating coefficients

(M1)

$$p = -7$$

(A1F)

$$q = 28$$

(A1F)

(b) (i) $r = 4, \theta = \frac{\pi}{3}$

B1,B1

2

(ii) $(2 + 2\sqrt{3}i)^n = \left(4e^{i\frac{\pi}{3}}\right)^n$

M1

$$= 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

AG

A1

2

(iii) $(2 - 2\sqrt{3}i)^n = 4^n \left(\cos \frac{n\pi}{3} - i \sin \frac{n\pi}{3} \right)$

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B1

$$\begin{aligned} \alpha^n + \beta^n + \gamma^n &= 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right) \\ &+ 4^n \left(\cos \frac{n\pi}{3} - i \sin \frac{n\pi}{3} \right) + \left(-\frac{1}{2} \right)^n \end{aligned}$$

M1

$$= 2^{2n+1} \cos \frac{n\pi}{3} + \left(-\frac{1}{2} \right)^n$$

AG

A1

3

[14]

Q6.

(a) Other root is $2 - 3i$

B1

1

- (b) Sum of roots = 4
ft error in (a)

B1F

So $b = -4$
ft wrong value for sum

B1F

Product is 13

B1

So $c = 13$
ft wrong value for product

B1F

Alternative:
Substituting $2 + 3i$ into equation

M1

Equating R and I parts

m1

$$12 + 3b = 0, \text{ so } b = -4$$

A1

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$$-5 + 2b + c = 0, \text{ so } c = 13$$

ft wrong value for b

A1F

4

[5]

Q7.

- (a) Use of $(\sum \alpha)^2 = \sum \alpha^2 + 2\sum \alpha\beta$

M1

$$1 = -5 + 2\sum \alpha\beta$$

A1

$$\sum \alpha\beta = 3$$

AG

A1
3

(b) $1(-5 - 3) = -23 - 3\alpha\beta\gamma$
For use of identity

M1

$$\alpha\beta\gamma = -5$$

A1
2

(c) $z^3 - z^2 + 3z + 5 = 0$

M1

For correct signs and “= 0”

A1F
2

(d) $\alpha^2 + \beta^2 + \gamma^2 < 0 \Rightarrow$ non real roots

B1

Coefficients real $\Rightarrow$ conjugate pair

B1
2

(e) $f(-1) = 0 \Rightarrow z + 1$ is a factor

M1A1

$$(z + 1)(z^2 - 2z + 5) = 0$$

A1

$$z = -1, 1 \pm 2i$$

A1
4

[13]

Q8.

(a) $2 + 3i$

B1
1

(b) (i) $\alpha\beta = 13$

B1
1

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha = 25$

M1A0 for -25 (no ft)

M1

$$\gamma(\alpha + \beta) = 12$$

A1F

$$\gamma = 3$$

ft error in $\alpha\beta$

A1F

3

Alternative

Attempt at $(z - 2 + 3i)(z - 2 - 3i)$

(M1)

$$z^2 - 4z + 13$$

(A1)

$$\text{cubic is } (z^2 - 4z + 13)(z - 3) \therefore \gamma = 3$$

(A1)

(3)

(iii) $p = -\sum \alpha = -7$

M1 for a correct method for either p or q

M1A1F

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$$q = -\alpha\beta\gamma = -39$$

ft from previous errors

p and q must be real

for sign errors in p and q allow M1 but A0

A1F

3

Alternative

Multiply out or pick out coefficients

(M1)

$$p = -7, q = -39$$

(A1,A1)

(3)

[8]

Q9.

- (a) (i) Roots are $\pm 4i$

M1 for one correct root or two correct factors

M1A1
2

- (ii) Roots are $1 \pm 4i$
M1 for correct method

M1A1
2

- (b) (i) $(1+x)^3 = 1+3x+3x^2+x^3$
M1A0 if one small error

M1A1
2

- (ii) $(1+i)^3 = 1+3i-3-i = -2+2i$
M1 if $i^2 = -1$ used

M1A1
2

- (iii) $(1+i)^3 + 2(1+i) - 4i$
with attempt to evaluate

M1

$$\dots = (-2+2i) + (2-2i) = 0$$

convincingly shown (AG)

A1

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[10]

Q10.

- (a) $\sum \alpha_i \beta_i = 6$

B1
1

- (b) (i) Sum of squares $< 0 \therefore$ not all real

E1

Coefficients real $\therefore$ conjugate pair

E1
2

- (ii) $\left(\sum \alpha_i\right)^2 = \sum \alpha_i^2 + 2\sum \alpha_i \beta_i$
A1 for numerical values inserted

M1A1

$$\left(\sum \alpha\right)^2 = 0$$

A1F

$$p = 0$$

cao

A1F

4

(c) (i) $-1 - 3i$ is a root

B1

Use of appropriate relationship eg $\sum \alpha = 0$
M0 if $\sum \alpha^2$ used unless the root 2 is checked

M1

Third root 2
incorrect p ✓

A1F

3

(ii) $q = -(-1 - 3i)(-1 + 3i)^2$
allow even if sign error

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M1

$= -20$
ft incorrect 3rd root

A1F

2

[12]

Q11.

(a) (i) $z^3 = \frac{4 \pm \sqrt{16 - 32}}{2}$

M1

$$= 2 \pm 2i$$

AG

A1

2

(ii) $2 + 2i = 2\sqrt{2}e^{\frac{\pi i}{4}}, 2 - 2i = 2\sqrt{2}e^{-\frac{\pi i}{4}}$

M1 for either result or for one of $r = 2\sqrt{2}, \theta = \pm \frac{\pi}{4}$

M1

$$\left(r = 2\sqrt{2} \quad \text{A1}, \quad \theta = \pm \frac{\pi}{4} \quad \text{A1} \right)$$

A1A1

$$z = \sqrt{2}e^{\frac{\pi i}{12} + \frac{2ki\pi}{3}} \text{ or } \sqrt{2}e^{\frac{-\pi i}{12} + \frac{2ki\pi}{3}}$$

M1 for either

M1

$$z = \sqrt{2}e^{\frac{\pm \pi i}{12}}, \sqrt{2}e^{\frac{\pm 3\pi i}{4}}, \sqrt{2}e^{\frac{\pm 7\pi i}{12}}$$

allow A1 for any 3 correct ft errors in $\pm \frac{\pi}{4}$

A2,1,0 F

6

(b) Multiplication of brackets

M1

Use of $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$

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A1

2

(c) $\left(z - \sqrt{2}e^{\frac{\pi i}{12}} \right) \left(z - \sqrt{2}e^{-\frac{\pi i}{12}} \right)$

$$= z^2 - 2\sqrt{2} \cos \frac{\pi}{12} z + 2$$

PI

M1A1F

$$\left(z^2 - 2\sqrt{2} \cos \frac{\pi}{12} z + 2 \right)$$

Product is $\left(z^2 - 2\sqrt{2} \cos \frac{7\pi}{12} z + 2 \right)$

(or $z^2 + 2z + 2$)

A1F

3

$$(z^2 - 2\sqrt{2} \cos \frac{3\pi}{4} z + 2)$$

[13]

Q12.

- (a) (i) Full expansion of product

M1

Use of $i^2 = -1$

m1

$$(2 + \sqrt{5}i)(\sqrt{5} - i) = 3\sqrt{5} + 3i$$

$\sqrt{5}\sqrt{5} = 5$ must be used – Accept not fully simplified

A1

3

(ii) $z^* = x - iy (= \sqrt{5} + i)$

M1

Hence result

Convincingly shown (AG)

A1

2

- (b) (i) Other root is $\sqrt{5} + i$

B1

1

- (ii) Sum of roots is $2\sqrt{5}$

B1

Product is 6

M1A1

3

- (iii) $p = -2\sqrt{5}, q = 6$

B1

ft wrong answers in (ii)

B1ft

2

[11]

Q13.

(a) $p = -4$

B1

$$(\alpha + \beta + \gamma)^2 = \Sigma \alpha^2 + 2\Sigma \alpha\beta$$

M1

$$16 = 20 + 2\Sigma \alpha\beta$$

A1

$$\Sigma \alpha\beta = -2$$

A1F

$$q = -2$$

A1F

5

(b) $3 - i$ is a root

B1

Third root is -2

B1F

$$\alpha\beta\gamma = (3 + i)(3 - i)(-2)$$

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$$= -20$$

Real $\alpha\beta\gamma$

A1F

$$r = +20$$

Real r

A1F

5

Alternative to (b)

Substitute $3 + i$ into equation

M1

$$(3 + i)^2 = 8 + 6i$$

B1

$$(3 + i)^3 = 18 + 26i$$

B1

$$r = 20$$

Provided r is real

A2,1,0

[10]



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Examiner reports

Q1.

This question was well answered with almost every student identifying the complex conjugate as a root of the equation. The methods for finding the third root were many and varied, as were the methods for finding the value of m . However, the majority of attempts were successful. Some students opted to find the three factors, expand the brackets and then compare coefficients. Although this is a perfectly valid method, it was quite time consuming, and prone to error.

A few students found m whilst calculating the third root. Many of these then wasted time in rewriting their method in part (b). Some of the less successful attempts confused factors with roots.

Q3.

This question which tested the complex numbers section of the specification proved, as expected, to be more demanding than the corresponding questions in recent MFP1 papers. In part (a)(i) a common wrong answer for $(z - 2i)^*$ was $x + iy + 2i$. Those students who started by writing $(x + iy - 2i)^*$ were generally more successful. There were some very good solutions to part (a)(ii), but some able students, having found the correct values for x and y , either stopped at that point or else gave their final answer as $\frac{1}{3} + \frac{4}{3}i$ which indicated that they thought that solving the given equation meant finding the value of $(z - 2i)^*$ and not the value of z . It was quite common to see some students making no attempt to equate real parts and imaginary parts, with final answers left in a form similar to $x + 4y - 3 + (2 - y - 4x)i$. The final part of the question was answered well by those who based their statement on:

- coefficients must be real for conjugate roots or
- sum of roots $(= 2p)$ cannot be $-10i$ if p is real or
- product of the roots $(p^2 + q^2)$ cannot be -29 if p and q are real.

However, correct statements were in the minority. Common wrong answers for example, 'it is not the difference of two squares' suggested that students were treating the roots $p \pm iq$ as the factors of the quadratic expression $z^2 + 10iz - 29$.

Q4.

Part (a) was invariably correctly quoted. However, in part (b) responses fell into several categories. The most successful students used the fact that α , β , and γ were each roots of the given cubic equation and therefore each satisfied that equation. Others quoted the formula for $\sum \alpha^3$ and, provided this quotation was correct, usually went on to secure full marks. A third category was those students who attempted to obtain $\sum \alpha^3$ by considering $(\sum \alpha)^3$ but these rarely succeeded, finding the algebra involved too complicated to handle. Some solutions ended at this juncture due to students not realising that a cubic equation with real coefficients must have a pair of complex conjugate roots if one root is complex. Part (c), apart from the odd slip of sign, was well done. In part (d), those students who replaced z by $\frac{1}{z}$ in the original cubic equation completed this part of the question quickly,

but those who tried to find the sums and products of the reciprocals of α , β and γ usually ended up with at least one incorrect result.

Q5.

This question provided a good source of marks for many candidates. The commonest error in part(a) was to write $\alpha\beta\gamma$ as -16 , overlooking the fact that the coefficient of x^3 was not unity and, of course, leading to an incorrect value for γ . This error perpetuated in part (a)(ii) with $\alpha + \beta + \gamma$ written as p or $-p$ and the same for q .

Parts (b)(i) and (b)(ii) were generally well done, although it should be stated that when answers are printed, candidates are expected to provide sufficient detail to show clearly how their answers are arrived at. Part (b)(iii) was also quite well done and it was pleasing

to note that in some cases where candidates had not arrived at $\gamma = -\frac{1}{2}$, they went back to part (a)(ii) to identify their error. Just occasionally in part (b)(iii), some candidates made

blatent errors in their attempt to convert $4^x \cos \frac{22\pi}{3} + 4^x \cos \frac{22\pi}{3}$ into $2^{2x+1} \cos \frac{22\pi}{3}$.

Q6.

Almost every candidate gave the right answer to part (a), and many went on to use the relationships between the roots and coefficients of a quadratic equation to answer part (b) correctly, though quite a number did not insert a minus sign for the coefficient of x . Some candidates followed other approaches, such as substituting the known root into the equation, expanding and equating real and imaginary parts, which if done carefully usually led to the right answers.

Q7.

Parts (a), (b) and (c) of this question were well done apart from odd sign errors here and there. However, there was much woolly thinking in part (d). For instance, it was not uncommon to see statements such as 'the cubic equation has real coefficients so it must have one real root and a conjugate pair of non-real roots' or 'since $\alpha^2 + \beta^2 + \gamma^2 = -5$, two of α , β and γ must be non-real'. Part (e) was poorly answered: it just did not seem to occur to candidates to use the factor theorem to find the real root. Instead they tried to use the symmetric relations between the roots in order to find them. This in turn led to heavy algebra with final abandonment.

Q8.

Responses to this question were good, and the vast majority of candidates produced a completely correct solution. If errors did occur they were usually arithmetic, although

occasionally p and q were given as $\sum \alpha$ and $\alpha\beta\gamma$ respectively with no consideration being given to their sign.

Q9.

This opening question gave almost all the candidates the opportunity to score a high number of marks. Even when careless errors were made, for example the omission of the plus-or-minus symbol in one or both sections of part (a), there was much correct work for the examiners to reward. The expansion of the cube of a binomial expression in part (b)(i) seemed to be tackled more confidently than in the past. Almost all the candidates used i^2

$= -1$ in part (b)(ii), though i^3 some were unsure how to deal with.

Q10.

Whilst part (a) was usually correctly done, part (b)(i) was poorly answered. Some candidates were able to comment on the condition that as the sum of the squares of the roots was less than zero there would have to be complex roots, but few stated the conditions that the coefficients of the cubic equation were all real. The value of p in part (b)(ii) was very often correct but in part (c)(i) a very common error as to use

$\sum \alpha^2 = -12$ in order to find the third root. This method led to $\alpha^2 = 4$ from which almost all candidates using this method wrote $\alpha = 2$ without even considering the possibility that α could equal -2 . Part (c)(ii) was usually worked correctly although $\alpha\beta\gamma = +q$ appeared from time to time.

Q11.

Candidates were usually able to establish the result in part (a) although the methods used were sometimes somewhat inelegant. Part (a)(ii) was reasonably well done although some carelessness was in evidence in this part. For instance, some candidates although

showing that the argument of z^3 was $\pm \frac{1}{4} \pi$ continued their solution with only $\pm \frac{1}{4} \pi$ and so arrived at a total of three roots. Others having reached $|z^3| = \sqrt{8}$ then thought that $|z| = \sqrt{8}$ also.

A few candidates used a method which, although possible, was not really suitable. They replaced the z^3 in $z^6 - 4z^3 + 8 = 0$ with $2 \pm 2i$ and so arrived at $z^6 = \pm 8i$. This latter equation gave the twelve roots of $z^{12} = -64$ and the method was incomplete unless 6 of the roots were rejected. Part (b) was generally well done, but part (c) was really only completed by candidates who had correctly answered part (a)(ii).

Q12.

Part (a)(i) was answered well by the majority of candidates, though the fourth term in the expansion often appeared as $-i^2$ instead of $-i^2 \sqrt{5}$, and $-i^2$ was frequently simplified to -1 in the next line.

The next part, (a)(ii), tested the candidates' knowledge of technique to some extent. Those who understood what they were being asked to do invariably realised that the necessary hard work had already been done in part (a)(i), so that very little writing was needed to pick up two further marks. Those who thought they had to prove, rather than verify, the given statement were awarded one mark for showing a knowledge of the conjugate of z , but then spent much valuable time trying to solve the equation. There were three indications that this was not what the question required: the word 'Hence', the word 'verify', and the award of only two marks.

Part (b) of this question was generally well answered, though some candidates did not seem to be aware of the implications of the statement that the coefficients in the quadratic equation were real. The answers to part (b)(ii) were sometimes given in terms of p and q , and in part (b)(iii) many candidates equated p to the sum of the roots instead of to this number preceded by a minus sign.

Q13.

Candidates were also able to achieve good results on this question. In part (a), where errors occurred they were almost always errors of signs. For instance the value of p was given as 2 instead of -2 or the formula for $(\sum \alpha)^2$ was incorrectly quoted as $\sum \alpha^2 - 2\sum \alpha\beta$. There were fewer completely correct solutions to part (b) often due to inelegant methods of solution. The most successful candidates obtained the values of the other two roots and then worked out the product $\alpha\beta\gamma$. The main loss of marks using this method was to equate r to $\alpha\beta\gamma$ instead of $-\alpha\beta\gamma$. The other main method of approach to this part of the question was to substitute $3 + i$ into the cubic equation with the values of p and q already found in part (a). However any error in the values of p and q or in the substitution inevitably led to r having a complex value. Surprisingly this did not seem to worry the candidates in spite of the fact that the question stated that r was real.



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